

Semiconductors and the World's Energy Crisis

Powerful, low-cost microcontrollers have enabled the traditional power-wasting universal motor in domestic appliances to be gradually replaced with efficient brushless motor. These brushless motors are typically 30 per cent more energy efficient. If all motors were replaced with brushless motors universally, this alone could save up to 50 TWh by 2020! And there are many other ways



VIVEK SHARMA
REGIONAL VICE PRESIDENT, GREATER
CHINA AND SOUTH ASIA REGION -
INDIA OPERATIONS, AND DIRECTOR,
STMICROELECTRONICS INDIA DESIGN
CENTERS

Worldwide demand for electrical energy is increasing and it looks like it will increase for the foreseeable future. Exxon Mobil (Outlook 2010) predicts that by 2030 global electricity demand will be 50 per cent higher than today's demand. Even if we found new reserves of fossils fuels, we would not be able to continue burning them because of their contribution to global warming.

"Unstoppable force meets immovable object" is a good metaphor to describe the situation. The 'unstoppable force' is the increasing

worldwide demand for energy. The 'immovable object' is the world's diminishing fossil-fuel reserves and the growing acceptance that burning these fuels is causing climate changes that will have profoundly negative effects on future generations.

Given the scale and complexity of the challenge, everyone must contribute to the solution. The semiconductor industry, in particular, can make the unstoppable force less unstoppable while making the immovable object more movable.

The dramatic increase in the number of people around the world using power and the number of power-hungry devices calls for a big change in the world's approach to electrical power. It may be unreasonable to expect to reduce the world's demand for electricity but we must aim to put the brakes on the rate of increase.

Two paths offer hope. The first approach is to reduce the power each product consumes and to make the generation and distribution of electrical energy more efficient. The second is to change consumers' energy-consumption patterns. Semiconductor technology can contribute to both.

In all electric and electronic product technologies, circuits and systems are developed by focusing on price vs performance. But now it is important to also consider power consumption. In fact, the International Energy Agency (IEA) expects

that increased efficiency in end-use products will be the most important contributor to the reduction of global energy-related carbon dioxide emissions, and better power efficiency could account for more than half of the total savings by the end of next decade.

There are several ways in which the energy consumption of end-use equipment can be reduced. One is via technology evolution. Replacing previous-generation transistors with state-of-the-art devices would contribute to saving 4 to 5 terrawatt-hour (TWh) per year in power-supply applications, assuming shipments of two billion units a year. This is equivalent to the energy produced by two 500MW nuclear power plants every year.

Even more important is the adoption of 'smart systems,' which combine multiple functions such as sensors, digital processing, connectivity and actuators. The effectiveness of this approach has already been proven in domestic appliances. There, the availability of powerful, low-cost microcontrollers has enabled the traditional power-wasting universal motor to be gradually replaced with efficient brushless motor. These brushless motors are typically 30 per cent more energy efficient. If all motors were replaced with brushless motors universally, this could save up to 50 TWh by 2020.

The second path, i.e., helping consumers change their energy-